Applied Economics

Multiple Linear Regression Model

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Winter 2025/26

Introduction

- From previous lecture.
  - A correlation between a "treatment" and an "outcome" does not neccessarily imply causation, because of **endogeneity**.
  - To adress endogeneity:
    - Randomize treatment: randomized control trials, natural experiments.
    - When randomization is not feasible, control for endogeneity using multiple linear regression model (MLR).
- Plan for today's lecture:
  1. Motivating the use of the MLR
  2. Interpreting the Parameters
  3. Deriving the OLS estimator
  4. Empirical Example in R

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Motivating Example for the use of the MLR in Economics

Underlying Economic Model

Economic question: does a higher school budget cause better student outcomes?

$$Y = \beta_0 + \beta_1 X_1$$

- Outcome variable $Y \in [0, 1]$: share of students that pass math exam.[1]
- Regressor $X_1 \in (0, \infty)$: school's per-student budget.[2]

Model assumptions are:

1. Single, representative school.
2. Model **assumes** a linear functional form between X_1 and Y.
3. Model **ignores** other factors impacting Y, such as student background.

[1] also known as explained variable, regressand, outcome

[2] also known as explanatory/independent variables, covariates

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Causal Effect of Expenditure on Education

Economic question: does school budget X_1 cause better student outcomes Y?

$$Y = \beta_0 + \beta_1 X_1$$

Let's compare our representative school's math score outcome Y when:

- School has a baseline budget $X_1 = A \in (0, \infty)$.
- We increase the baseline budget in one unit: $X_1 = A + 1$

$$E[Y \mid X = A + 1] - E[Y \mid X = A] = \beta_1$$

Causal Effect

β_1 represents how much a $+1$ unit increase in the per-student budget causes the school's share of good students to change.

Causal Effect of Expenditure on Education

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Equivalent Statistical Model

Our aim is to estimate the β_1 causal parameter.

Equivalent statistical model, for the whole school population N:

$$Y = \beta_0 + \beta_1 X_1 \quad \rightarrow \quad Y_i = \beta_0 + \beta_1 X_{1i} + u_i, \quad i = 1, \ldots, N$$

1. Representative schoool $\rightarrow$ Population of N schools, indexed by i.
2. Linear functional form assumption is preserved.
3. Error term u_i encompasses all the variables that affect Y_i, other than X_{1i}.

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Causal Effect in the Statistical Model

$$Y = \beta_0 + \beta_1 X \quad \rightarrow \quad Y_i = \beta_0 + \beta_1 X_{1i} + u_i$$

Original causal effect becomes into **Average** Causal effect:

$$E[Y \mid X = A + 1] - E[Y \mid X = A] = \beta_1 \quad \rightarrow \quad E[Y_i \mid X_{1i} = A + 1] - E[Y_i \mid X_{1i} = A]$$

Average Causal Effect

Features the original β_1 and an additional term.

$$E[Y_i|X_{1i} = A + 1] - E[Y_i|X_{1i} = A] = \beta_1 + E[u_i|X_{1i} = A + 1] - E[u_i|X_{1i} = A]$$

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Mean independence of u_i from X_{1i}

For β_1 to still represent the average **causal** effect, we need to assume that the additional term is zero.

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Mean Independence Assumption

$$E[u_i|X_{1i} = A + 1] = E[u_i|X_{1i} = A]$$

All the underlying characteristics u_i that determine the outcome Y_i should be the same between the treated and the non-treated units.

Does this assumption make sense in our context? Examples?

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Motivation for Use of MLR

How can we **tame** endogeneity, make **more credible** causal claims? Enrich our model with k multiple regressors.

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i \quad \rightarrow \quad Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

- Original u_i encompasses all the variables that affect Y_i, other than X_{1i}.
- New u_i encompasses all the variables that affect Y_i, other than $X_{1i}, X_{2i}, ..., X_{ki}$.

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Causality in MLR

Conditional Mean Independence Assumption

$$E[u_i|X_i = A + 1, X_{2i}, ..., X_{ki}] = E[u_i|X_i = A, X_{2i}, ..., X_{ki}]$$

To claim causality, all the underlying characteristics u_i that determine the outcome Y_i, other than $X_{2i}, ..., X_{ki}$ should be the same between the treated and the non-treated units.

- Weaker assumption than previous Mean Independance.
- By adding all relevant K covariantes, bias dissappears $\rightarrow$ Can claim causality!

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Interpreting the Regression Parameters

Conditional Expectation Function

Conditional Expectation Function (CEF)
Is a function that takes some regressor values $X_{1i}, X_{2i}, ..., X_{ki}$, then returns the corresponding expected value of dependent variable Y_i

$$CEF(X_{1i}, X_{2i}, ..., X_{ki}) = E[Y_i|X_{1i}, X_{2i}, ..., X_{ki}] = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki}$$

What does β_j mean?

1. Starting with the CEF for some regressor values $CEF(X_{1i}, ..., X_{ji}, ..., X_{ki})$.
2. Increase X_j by one unit: $CEF(X_{1i}, ..., X_{ji} + 1, ..., X_{ki})$.
3. $\beta_j = CEF(X_{1i}, ..., X_{ji} + 1, ..., X_{ki}) - CEF(X_{1i}, ..., X_{ji}, ..., X_{ki})$

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Average Partial Effects

$$\beta_j = CEF(X_{1i}, ..., X_{ji} + 1, ..., X_{ki}) - CEF(X_{1i}, ..., X_{ji}, ..., X_{ki})$$

Average Partial Effect

- β_j represents the average partial effect that increasing X_j by one unit has on outcome Y...
- all while keeping rest of covariates constant (as in randomized control trials).
- Equivalent to Ceteribus paribus concept.

Be cautious with language implying causal relationships.

- β_j will always represent a **correlation**...
- but will only represent a partial **causal** effect if the conditional mean independence assumption holds.

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The OLS Estimator

Parameters, Estimates, and Estimators

So far, we focused on **population** parameters β. Remember! We only observe a $n < N$ iid sample of the population.

- Parameter (β) is an **unknown, deterministic number** that describes relationships between two variables in the population.
- Estimate ($b, \hat{\beta}$) is a **random variable** that approximates the true parameter based on the observed dataset.
- Estimator (OLS) is a **function** that takes the observed and returns an estimate. Different population draws yields different estimates.

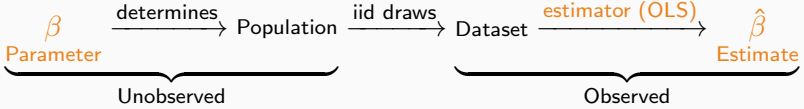
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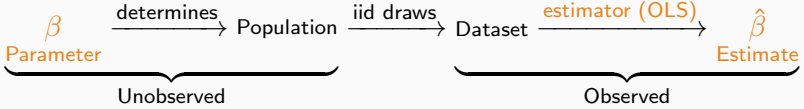
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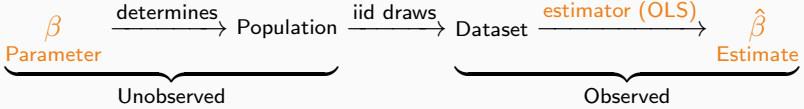
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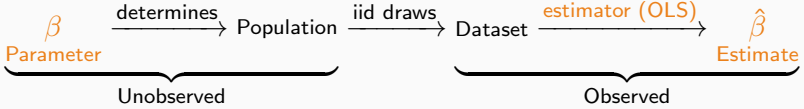
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From Population to Data

Statistical model for the unobserved population:

- Outcome, regressors are **random variables** with joint distribution $(Y, X_1, \ldots, X_k)$.
- There is a population of size N we (almost) never get to fully observe.
- Error term u_i

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i \quad i = 1, ..., N$$

Statistical model for the observed data:

- We observe $n < N$ iid draws from the joint distribution.
- Dataset with n observations (rows) and $k + 1$ variables (columns):
- Residual $\hat{u}_i$

$$Y_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \cdots + b_k X_{ki} + \hat{u}_i \quad i = 1, ..., n$$

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Writing the MLR model with vector and matrix notation

1. A dataset with n observations and k covariates can be divided into:
   - 1.1 $\underset{(n\times 1)}{\mathbf{Y}}$ vector for grouping the outcome variable of all observations.
   - 1.2 $\underset{(n\times k+1)}{\mathbf{X}}$ matrix grouping all the covariates and constant term, for all observations.
2. $\underset{(k+1\times 1)}{\mathbf{b}}$ vector of parameter **estimates**, including constant term b_0.
3. $\underset{(n\times 1)}{\hat{\mathbf{u}}}$ vector stacking the residuals for all observations.

As a result, our MLR model for observed data is equivalent to a **sample** model:

$$Y_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \cdots + b_k X_{ki} + \hat{u}_i \quad \rightarrow \quad \mathbf{Y} = \mathbf{X} \times \mathbf{b} + \hat{\mathbf{u}}$$

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As a result, our MLR model for observed data is equivalent to a **sample** model:

$$Y_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + \cdots + b_k X_{ki} + \hat{u}_i \quad \rightarrow \quad \mathbf{Y} = \mathbf{X} \times \mathbf{b} + \hat{\mathbf{u}}$$

Writing the MLR model with vector and matrix notation

1. A dataset with n observations and k covariates can be divided into:
   - 1.1 $\underset{(n\times 1)}{\mathbf{Y}}$ vector for grouping the outcome variable of all observations.
   - 1.2 $\underset{(n\times k+1)}{\mathbf{X}}$ matrix grouping all the covariates and constant term, for all observations.
2. $\underset{(k+1\times 1)}{\mathbf{b}}$ vector of parameter **estimates**, including constant term b_0.
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How can we estimate β?

1. Start with vector of candidate parameter estimates $\underset{(k\times 1)}{\mathbf{b}}$
2. Compute the vector of **residuals**:

$$\underset{(n\times 1)}{\hat{\mathbf{u}}} = \underset{(n\times 1)}{\mathbf{Y}} - \underset{(n\times k)}{\mathbf{X}}' \times \underset{(k\times 1)}{\mathbf{b}}$$

3. **Sum of Squared Residuals (SSR):** measures the overall discrepancy between the observed and predicted outcome variable:

$$\underset{(1\times 1)}{SSR(b)} = \underset{(1\times n)}{\hat{\mathbf{u}}}' \times \underset{(n\times 1)}{\hat{\mathbf{u}}}$$

4. The Ordinary Least Squares (OLS) estimator Finds the vector of parameters b that minimize $SSR(b)$. How?

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OLS Estimator Formula

Rewrite the SSR(b) objective function:

$$\begin{aligned} SSR(b) &= \hat{\mathbf{u}}'\hat{\mathbf{u}} \\ &= (\mathbf{Y} - \mathbf{X}\boldsymbol{b})'(\mathbf{Y} - \mathbf{X}\boldsymbol{b}) \\ &= \mathbf{Y}'\mathbf{Y} - \boldsymbol{b}'\mathbf{X}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\boldsymbol{b} + \boldsymbol{b}'\mathbf{X}'\mathbf{X}\boldsymbol{b} \\ &= \mathbf{Y}'\mathbf{Y} - 2\boldsymbol{b}'\mathbf{X}'\mathbf{Y} + \boldsymbol{b}'\mathbf{X}'\mathbf{X}\boldsymbol{b} \end{aligned}$$

To find the minimum, first-order condition with respect to vector $\boldsymbol{b}$:

$$\frac{\partial(\hat{\mathbf{u}}'\hat{\mathbf{u}})}{\partial \boldsymbol{b}} = -2\mathbf{X}'\mathbf{Y} + 2(\mathbf{X}'\mathbf{X})\boldsymbol{b} = 0$$

Solving the first-order condition for $\mathbf{b}$ yields the OLS estimator:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

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Empirical Example

How does public spending on education affect students' performance?

For a dataset of $n =$ schools, we observe:

$Math_i$ Percentage share of fourth graders with satisfactory math performance in school i

$Expenditures_i$ Expenditures per student (in \$1000) in school i

We propose the following model to test the effectiveness of eduational spending on student performance:

$$Math_i = \beta_0 + \beta_1 Expenditures_i + u_i$$

Open w2_code.R

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Data Preparation

```r
# Open dataset
df = read.csv("testscoresMI.csv")
# Create expenditures per student variable
df$expend_ps = df$expend/df$enroll
df$expend_ps_thous = df$expend_ps/1000
# Create constant variable
df$constant = 1
```

```r
# Dataset in matrix notation
X <- as.matrix(df[,names(df) %in% c("constant","expend_ps_thous")])
Y <- as.matrix(df[,names(df) %in% c("math4")])
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Single Linear Regression in R

```r
# Single linear regression using OLS formula:
beta_hat_m1 = solve(t(X) %*% X) %*% t(X) %*% Y
```

```r
# Linear regression using R function lm(), store results as slr_math
slr_math = lm(math4 ~ expend_ps_thous, data = df)
# Summarize regression results:
summary(slr_math)
```

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Inspecting First Regression Results

Why do we find negative coefficient $\beta_1 < 0$ on the spending variable for math (and reading) performance?

$$Math_i = \beta_0 + \beta_1 Expenditures_i + u_i$$

Omitted variable bias: Other relevant factors influencing student performance (in the error term) are likely correlated with expenditures. Which?

One hypothesis:

- Government spends more money on schools with a lot of disadvantaged students.
- Disadvantaged students have less means to learn math.

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Multiple Linar Regression in R

To control for the share of disadvantaged students in the school, we add the share of students eligible for free lunch $lunch_i \in [0, 1]$ to the regression:

$$Math_i = \beta_0 + \beta_1 Expenditures_i + \beta_2 lunch_i + u_i$$

Implementation in R:

```
mlr_math = lm(math4 ~ expend_ps_thous + lunch, data = df)
summary(mlr_math)
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Implementation in R:

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mlr_math = lm(math4 ~ expend_ps_thous + lunch, data = df)
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```

Reporting Regression Results: LaTeX Table

```
install.packages("stargazer") # if not installed
library(stargazer)
stargazer(slr_math, mlr_math, type = "text")
```

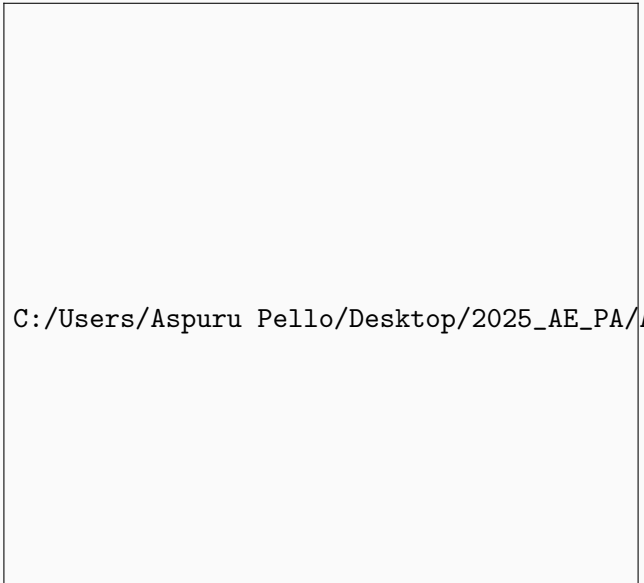
Reporting Regression Results: LaTeX Table

	Dependent variable:	
	math4	
	(1)	(2)
expend_ps_thous	−0.570	2.083***
	(0.428)	(0.350)
lunch		−0.471***
		(0.014)
Constant	74.870***	79.568***
	(2.272)	(1.813)
Observations	1,823	1,823
R^2	0.001	0.368
Note:	*p<0.1; **p<0.05; ***p<0.01	

Reporting Regression Results: Plot

```
install.packages("jtools") # if not installed
library(jtools)
plot_coefs(slr_math, mlr_math,
           coefs = c("Expenditures per student (in $1000)" =
                       "expend_ps_thous"))
```

Reporting Regression Results: Plot



Recap

In an empirical analysis, our goal is often to estimate a causal relationship between an explanatory variable X_1 and an outcome Y. First attempt:

$$Y_i = \beta_0 + \beta_1 X_1 + u_i$$

Extremely common problem: X_1 is correlated with u_i

$\rightarrow$ as X_1 changes, also other determinants of Y change. Thus, subsequent change in Y does not reflect the causal effect of X_1. Remember: We say that X_1 causes Y when if we were to intervene and change the value of X_1 without changing anything else then Y would also change as a result.

$\Rightarrow$ Simple regression model recovers a correlation, but not a causal effect.

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Recap

We can use the multiple linear regression model to keep everything else (that we can observe) fixed

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_k X_k + u_i$$

Now, $\hat{\beta}_1$ reflects the average change in Y given a one unit increase in X_1, while holding X_2, ..., X_k fixed.

$\rightarrow$ gets us closer to a randomised treatment allocation.

If there are no factors left in u_i that are correlated with the regressors, i.e. $E[U_i|X_1, ..., X_k] = 0$, then $\hat{\beta}_1$ estimates the causal impact of X_1 on Y.

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Recommended Literature

Heiß Chapter 3 - 3.3